Graphs

Graphs and Graph Models

Definition 1: A *graph* $G = (V, E)$ consists of V, a nonempty set of *vertices* (or *nodes*) and E, a set of *edges*. Each edge has either one or two vertices associated with it, called its *endpoints*. An edge is said to *connect* its end points.

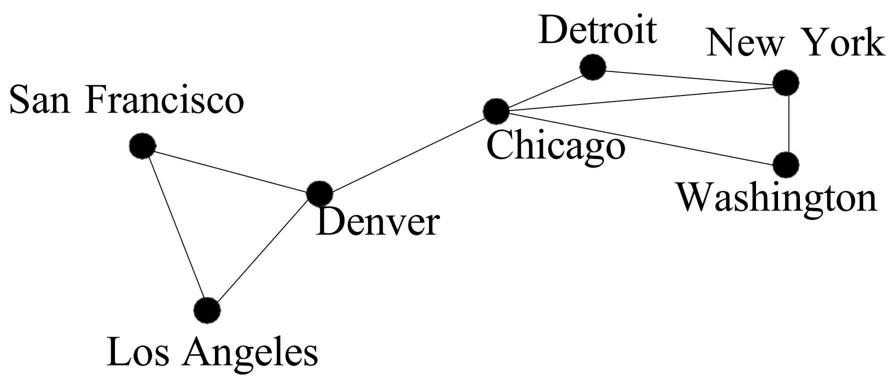


Figure: A computer network represented by a graph.

When there is an edge of a graph associated to $\{u, v\}$, we can say that $\{u, v\}$ is an edge of the graph.

If an edge connects a vertex to itself, it is called a **loop**.

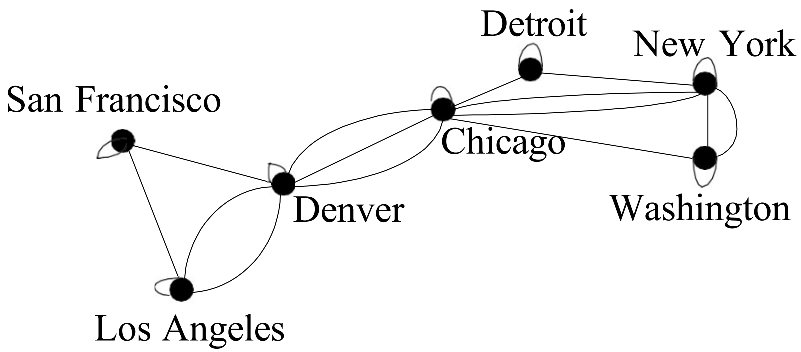


Figure: A graph representing a computer network with multiple edges and loops.

So far the graphs we have introduced are **undirected graphs**. Their edges are also said to be **undirected**.

Definition 2: A *directed graph* (or *digraph*) $G = (V, E)$ consists of a nonempty set of vertices V and a set of *directed edges* (or *arcs*) E. Each directed edge is associated with an ordered pair of vertices. The directed edge associated with the ordered pair (u, v) is said to *start* at u and *end* at v.

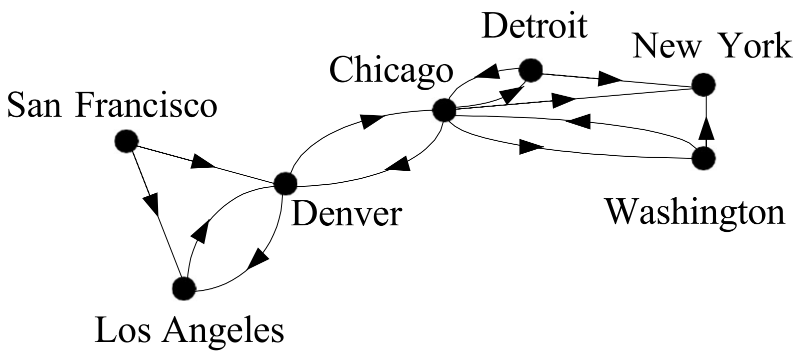


Figure: A graph representing a communication network with one-way communications links.

We use an arrow pointing from u to v to indicate the direction of the edge that starts at u and ends at v.

We call (u, v) an edge if there is an edge associated to it in the graph.

A graph with both directed and undirected edges is called a **mixed graph**.

Graph Terminology and Special Types of Graphs

Definition 1: Two vertices u and v in an undirected graph G are called *adjacent* (or *neighbors*) in G if u and v are endpoints of an edge of G. If e is associated with $\{u, v\}$, the edge e is called *incident with* the vertices u and v. The edge e is also said to *connect* u and v. The vertices u and v are called *endpoints* of an edge associated with $\{u, v\}$.

Definition 2: The *degree of a vertex in an undirected graph* is the number of edges incident with it, except that a loop at a vertex contributes twice to the degree of that vertex. The degree of the vertex v is denoted by $\deg(v)$.

Example: What are the degrees of the vertices in the graph G and H displayed in Figure 1?

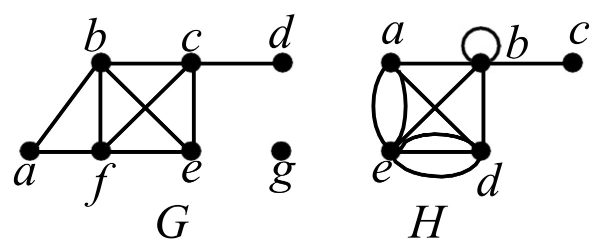


Figure: The Undirected Graphs G and H.

Solution:
In G, $\deg(a) = 2$, $\deg(b) = 4$, $\deg(c) = 4$, $\deg(d) = 1$, $\deg(e) = 3$, $\deg(f) = 4$, $\deg(g) = 0$

In H, $\deg(a) = 4$, $\deg(b) = 6$, $\deg(c) =1$, $\deg(d) = 5$, $\deg(e) = 6$

A vertex of degree zero is called **isolated**. A vertex of degree one is called **pendant**.

Theorem 1: THE HANDSHAKING THEOREM
Let $G = (V, E)$ be an undirected graph with $|E|$ edges. Then

$$2|E| = \sum_{v \in V} \deg(v)$$

Example: How many edges are there in a graph with 10 vertices each of degree six?

Solution:
Sum of degrees of the vertices = $6 \cdot 10 = 60$
$2|E| = 60$
$\therefore |E| = \frac{60}{2} = 30$

Definition 3: When (u, v) is an edge of the graph G with directed edges, u is said to be *adjacent to* v and v is said to be *adjacent from* u. The vertex u is called the *initial* or *start vertex* of (u, v), and v is called the *terminal* or *end vertex* of (u, v). The initial vertex and terminal vertex of a loop are the same.

Definition 4: In a graph with directed edges the *in-degree of a vertex v*, denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The *out-degree of v*, denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex. (Note that a loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.)

Example: Find the in-degree and out-degree of each vertex in the graph G with directed edges shown in Figure 2.

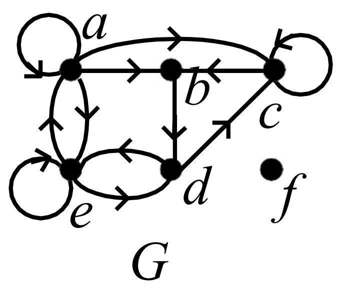


Figure: The Directed Graph G.

Solution:

The in-degree in G are: $\deg^-(a) = 2$, $\deg^-(b) = 2$, $\deg^-(c) = 3$, $\deg^-(d) = 2$, $\deg^-(e) =3$, $\deg^-(f) = 0$

The out-degree in G are: $\deg^+(a) = 4$, $\deg^+(b) = 1$, $\deg^+(c) = 2$, $\deg^+(d) = 2$, $\deg^+(e) =3$, $\deg^+(f) = 0$

Theorem 3: Let $G = (V, E)$ be a graph with directed edges. Then

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

Some Special Graphs

Example: Complete Graphs

The **complete graph on *n* vertices**, $n \geq 1$, denoted by K_n, is the graph that contains exactly one edge between each pair of distinct vertices.

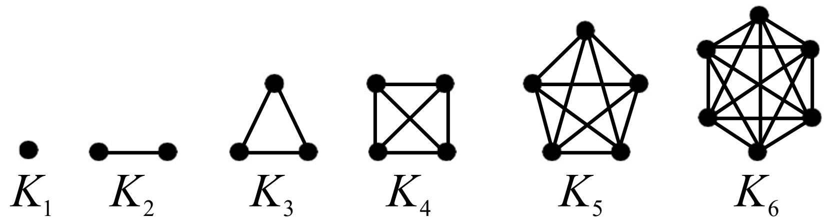


Figure: The Graphs K_n for $1 \leq n \leq 6$.

Example: Cycles

The **cycle** C_n, $n \geq 3$, consists of n vertices $v_1, v_2, \ldots, v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$.

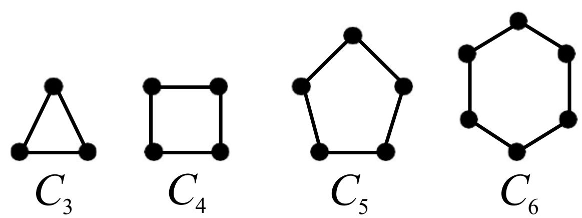


Figure: The Cycles C_3, C_4, C_5, and C_6.

Example: Wheels

We obtain the **wheel** W_n when we add an additional vertex to the cycle C_n, for $n \geq 3$, and connect this new vertex to each of the n vertices in C_n, by new edges.

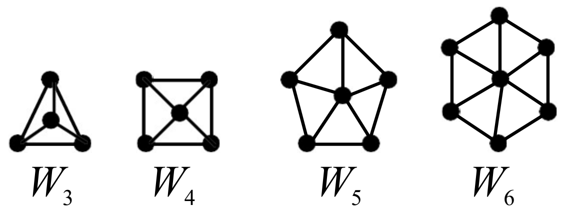


Figure: The Wheels W_3, W_4, W_5, and W_6.

Bipartite Graphs

Definition 5: A simple graph G is called *bipartite* if its vertex set V can be partioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2 (so that no edge in G connects either two vertices in V_1 or two vertices in V_2). When this condition holds, we call the pair (V_1, V_2) a *bipartition* of the vertex set V of G.

Example: C_6 is bipartite as shown in Figure 7.

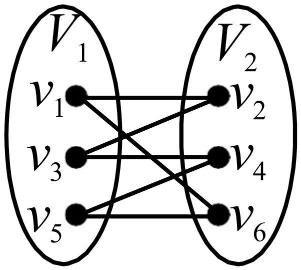
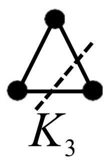


Figure: Showing that C_6 is Bipartite.

Example: K_3 is not bipartite.



Example. Are the graphs G and H displayed in Figure 8 bipartite?

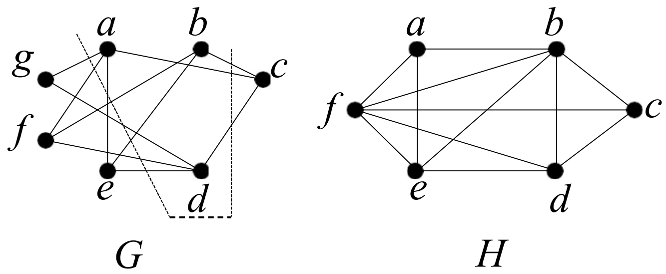


Figure:. The Undirected Graphs G and H.

Solution:
Graph G is bipartite, since its vertex set is the union of two disjoint subsets, $\{a, b, d\}$ and $\{c, e, f, g\}$, and each edge connects a vertex in one of these subsets to a vertex in the other subset.

Graph H is not bipartite, since its vertex set cannot be partitioned into two subsets so that edges do not connect two vertices from the same subset.

Representing Graphs and Graph Isomorphism

Representing Graphs

Adjacency Lists

One way to represent a graph is to use **adjacency lists**, which specify the vertices that are adjacent to each vertex of the graph.

Example: Use adjacency lists to describe the graph given in Figure 1.

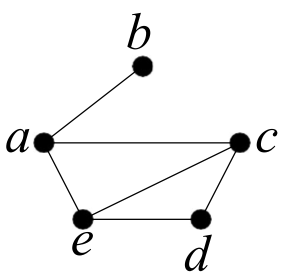


Figure:

Solution:

Table: An Edge List for a Graph.

Vertex	Adjacent vertices
a	*b, c, e*
b	*a*
c	*a, d, e*
d	*c, e*
e	*a, c, d*

Example: Represent the directed graph shown in Figure 2 by listing all the vertices that are the terminal vertices of edges starting at each vertex of the graph.

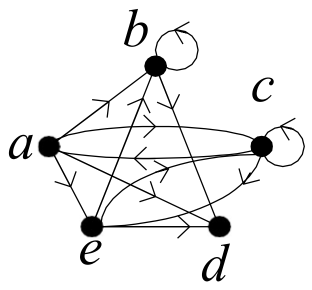


Figure:. A Directed Graph.

Solution:

Table 2. An Edge List for a Directed Graph.

Initial vertex	Terminal Veritces
a	*b, c, d, e*
b	*b, d*
c	*a, c, e*
d	
e	*b, c, d*

Adjacency Matrices

Suppose that $G = (V, E)$ is a graph where $|V| = n$. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \ldots, v_n$. The **adjacency matrix A** of G, with respect to this listing of the vertices, is an $n \times n$ zero-one matrix with 1 as its (i, j)th entry when v_i and v_j are adjacent, and 0 as its (i, j)th entry when they are not adjacent. In other words, if its adjacency matrix is $\mathbf{A} = [a_{ij}]$, then

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

Example: Use an adjacency matrix to represent the graphs shown in Figure 3.

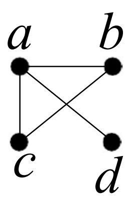


Figure: A Simple Graph.

Solution:

We order the vertices as a, b, c, d. The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Example: Draw a graph with the adjacency matrix

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

with respect to the ordering of vertices a, b, c, d.

Solution:

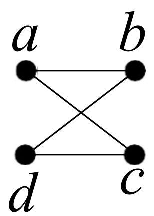


Figure:. A Graph with the Given Adjacency Matrix.

The matrix for a directed graph $G = (V, E)$ is $\mathbf{A} = [a_{ij}]$, where

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$

Example: Represent the directed graph shown in Figure 2 by adjacency matrix.

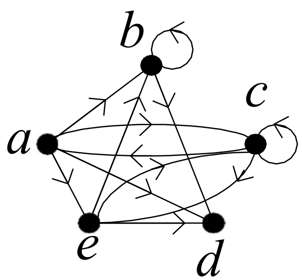


Figure: A Directed Graph.

Solution: We order the vertices as a, b, c, d, e. The matrix representing this graph is

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$